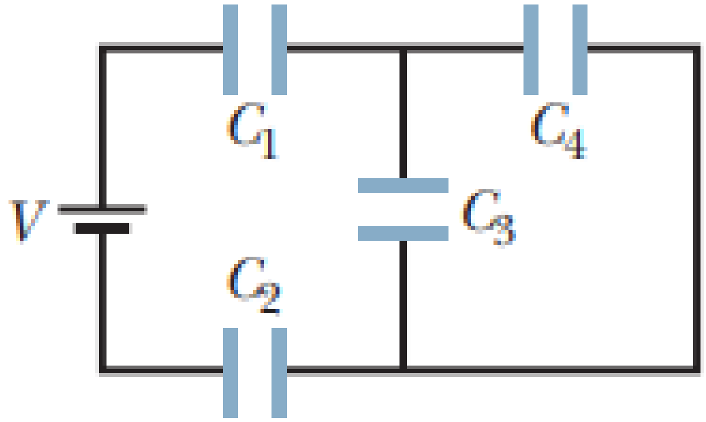
Lecture 12

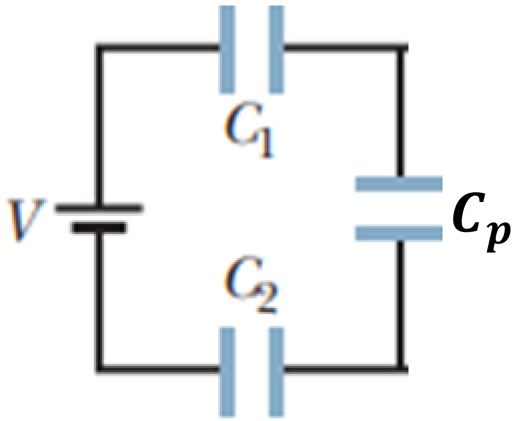
Chapter 25

Math Problems

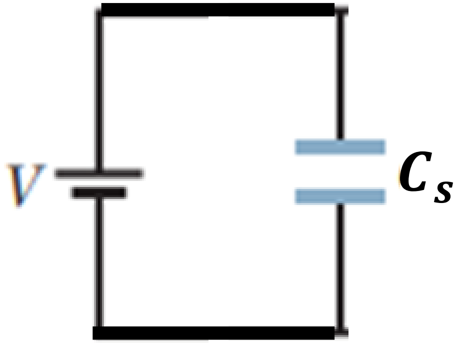
Q1. In Fig. , $V = 9.0\,V, C_1, C_2 = 30\ \mu F$, and $C_3, C_4 = 15\ \mu F$.
What is the charge on capacitor 4?



$$\boldsymbol{C_p = C_3 + C_4}$$
$$= 15 + 15$$
$$= 30\ \mu F$$



$$\boldsymbol{\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_p} + \frac{1}{C_2}}$$
$$= \frac{1}{30} + \frac{1}{30} + \frac{1}{30}$$
$$= \frac{3}{30}$$
$$= \frac{1}{10}$$
$$\boldsymbol{C_s} = 10\ \mu F$$



$$\boldsymbol{q_s = C_s V} = 10 \times 10^{-6} \times 9 = 90 \times 10^{-6} c$$

Therefore, $\boldsymbol{q_p = 90 \times 10^{-6} c}$

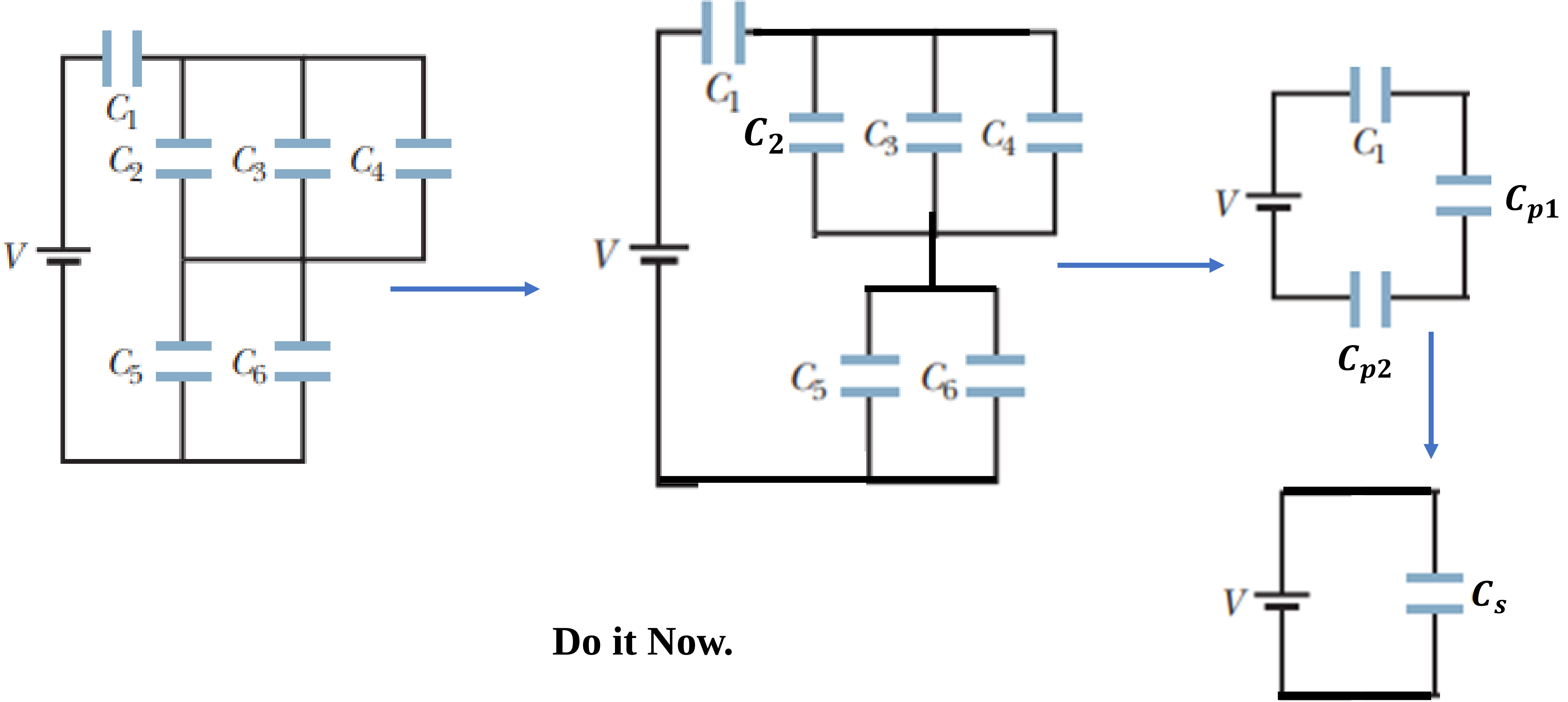
$$\boldsymbol{q_p = C_p V_p}$$

$$\boldsymbol{V_p = \frac{q_p}{C_p} = \frac{90 \times 10^{-6}}{30 \times 10^{-6}} = 3V}$$

Therefore, potential difference on $\boldsymbol{C_3}\, and\ \boldsymbol{C_4}$are $\boldsymbol{3V}$

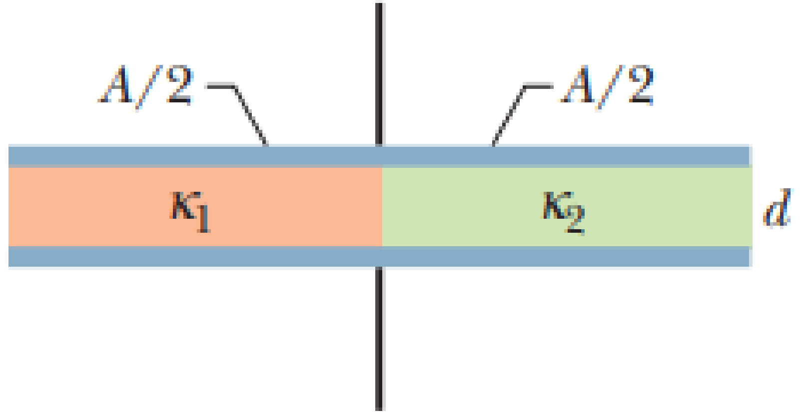
So, $\boldsymbol{q_4 = C_4 V_p = 15 \times 10^{-6} \times 3 = 45 \times 10^{-6} c = 45 \mu c}$

Q2. In Fig. , $V = 12\ V, C_1, C_5, C_6 = 6.0\ \mu F$, and $C_2, C_3, C_4 = 4.0\ \mu F$. What are (a) the net charge stored on the capacitors and (b) the charge on capacitor 4?



Do it Now.

Q3. Figure shows a parallel plate capacitor with a plate area $A = 5.56\ cm^2$ and separation $d = 5.56\ mm$. The left half of the gap is filled with material of dielectric constant $k_1 = 7.00$; the right half is filled with material of dielectric constant $k_2 = 12.0$ .What is the capacitance?



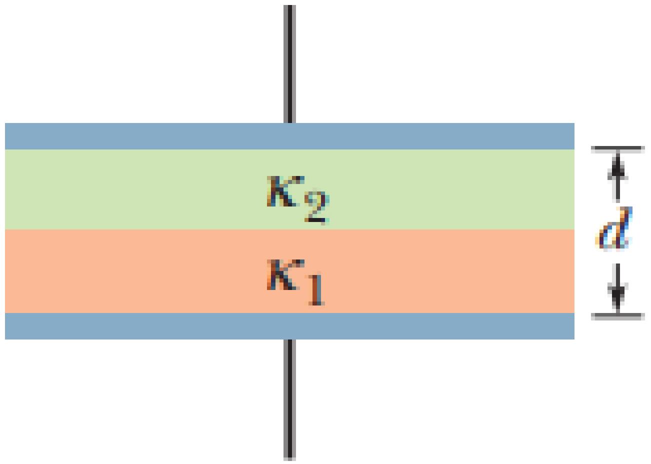
Let, the capacitance of capacitors of left side and right are C_1 and C_2

$$\boldsymbol{C_1} = \frac{\boldsymbol{\varepsilon_0 k_1 \frac{A}{2}}}{\boldsymbol{d}} = \frac{\boldsymbol{\varepsilon_0 k_1 A}}{\boldsymbol{2d}} = \frac{8.854 \times 10^{-12} \times 7 \times 5.56 \times 10^{-4}}{\boldsymbol{2} \times 5.56 \times 10^{-3}} = 3.1 \times 10^{-12} F$$

$$\boldsymbol{C_2} = \frac{\boldsymbol{\varepsilon_0 k_2 \frac{A}{2}}}{\boldsymbol{d}} = \frac{\boldsymbol{\varepsilon_0 k_2 A}}{\boldsymbol{2d}} = \frac{8.854 \times 10^{-12} \times 12 \times 5.56 \times 10^{-4}}{\boldsymbol{2} \times 5.56 \times 10^{-3}} = 5.31 \times 10^{-12} F$$

$$\boldsymbol{C_p} = \boldsymbol{C_1} + \boldsymbol{C_2} = 3.1 \times 10^{-12} + 5.31 \times 10^{-12} = 8.41 \times 10^{-12} F$$

Q4. Figure shows a parallel-plate capacitor with a plate area $A = 7.89\ cm^2$ and plate separation $d = 4.62\ mm$. The top half of the gap is filled with material of dielectric constant $k_1 = 11.0$; the bottom half is filled with material of dielectric constant $k_2 = 12.0$. What is the capacitance?



Let, the capacitance of capacitors of bottom and top zone are C_1 and C_2

$$\boldsymbol{C_1} = \frac{\boldsymbol{\varepsilon_0 k_1 A}}{\frac{\boldsymbol{d}}{\boldsymbol{2}}} = \frac{\boldsymbol{2\varepsilon_0 k_1 A}}{\boldsymbol{d}} = \frac{\boldsymbol{2} \times 8.854 \times 10^{-12} \times 11 \times 7.89 \times 10^{-4}}{4.62 \times 10^{-3}}$$

$$= 3.32 \times 10^{-11} F$$

$$\boldsymbol{C_2} = \frac{\boldsymbol{\varepsilon_0 k_1 A}}{\frac{\boldsymbol{d}}{\boldsymbol{2}}} = \frac{\boldsymbol{2\varepsilon_0 k_2 A}}{\boldsymbol{d}} = \frac{\boldsymbol{2} \times 8.854 \times 10^{-12} \times 12 \times 7.89 \times 10^{-4}}{4.62 \times 10^{-3}}$$

$$= 3.63 \times 10^{-11} F$$

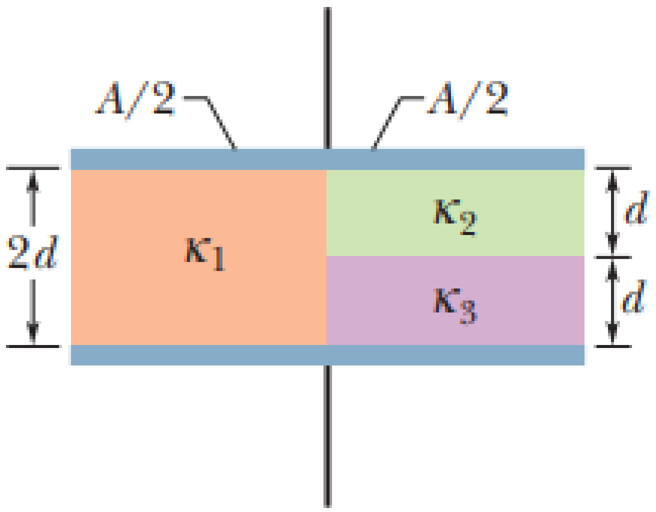
$$\frac{\boldsymbol{1}}{\boldsymbol{C_s}} = \frac{\boldsymbol{1}}{\boldsymbol{C_1}} + \frac{\boldsymbol{1}}{\boldsymbol{C_2}}$$

$$C_s = \frac{C_1 C_2}{C_1 + C_2}$$

$$= \frac{3.32 \times 10^{-11} \times 3.63 \times 10^{-11}}{3.32 \times 10^{-11} + 3.63 \times 10^{-11}}$$

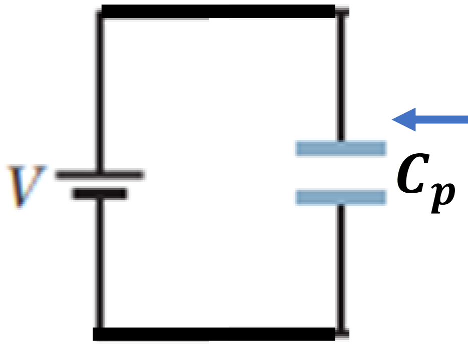
$$\boldsymbol{C_s} = 1.73 \times 10^{-11} F$$

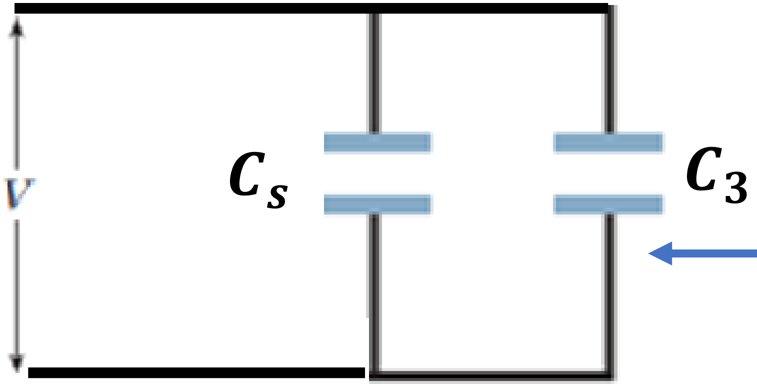
Q5. Figure shows a parallel-plate capacitor of plate area $A = 10.5\ cm^2$ and plate separation $2d = 7.12\ mm$. The left half of the gap is filled with material of dielectric constant $k_1 = 21.0$; the top of the right half is filled with material of dielectric constant $k_2 = 42.0$; the bottom of the right half is filled with material of dielectric constant $k_3 = 58.0$.What is the capacitance?

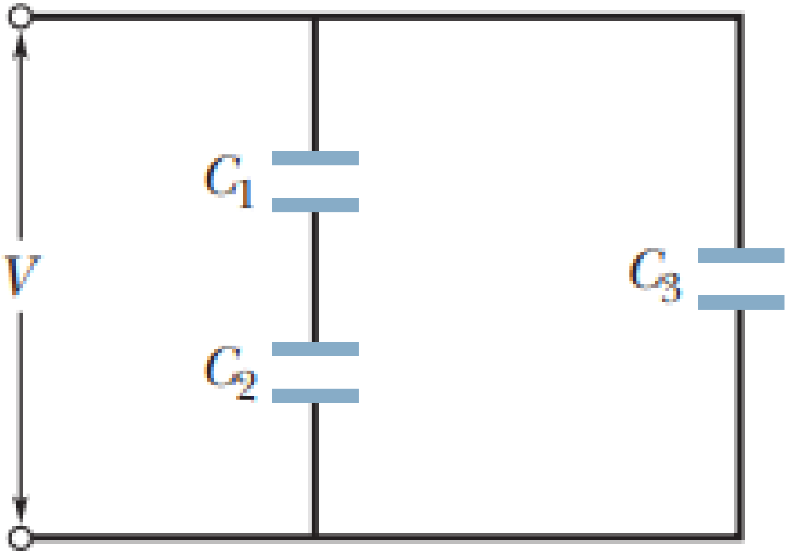


Do it Now

Q6. In Fig, a potential difference $V = 100\ V$ is applied across a capacitor arrangement with capacitances $C_1 = 10\ \mu F$, $C_2 = 5.0\ \mu F$, and $C_3 = 4.0\ \mu F$. What are (a) charge q_3, (b) potential difference V_3, and (c) stored energy U_3 for capacitor 3, (d) q_1, (e) V_1, and (f) U_1 for capacitor 1, and (g) q_2, (h) V_2, and (i) U_2 for capacitor 2?







$$\boldsymbol{C_p = C_3 + C_s}$$
$$= 4 + 3.33$$
$$= 7.33\ \mu F$$

$$\boldsymbol{\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}}$$

$$C_s = \frac{10 \times 5}{10 + 5} = 3.33\ \mu F$$

Do it Now

Thank You